

Capacitance and Dielectrics

Chapter 25, RHW 10th edition.

Energy Stored in An Electric field

① Work must be done by an external agent to charge a Capacitor.

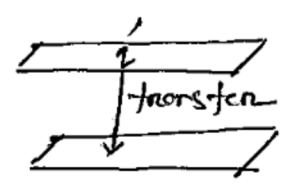
② We can imagine doing the work ourselves by transferring electrons from one plate to other, one by one.

③ As the charges build, so does the electric field between the plates which opposes the continued transfer.

⚡ So greater amounts of work are required, actually a battery does all this work for us.

We visualize the work as being stored as electric potential energy in the electric field between the plates.

At a given instant, a charge q' has been transfer from one plate of capacitor to the other. The potential difference V' between the plates at that instant will be q'/C . If an extra increment of charge dq' is then transferred, the increment of the work required will be,



$$dw = V' dq' = \frac{q'}{C} dq'$$

$$w = \int dw = \frac{1}{C} \int_0^q q' dq' = \frac{q^2}{2C}$$

$$U = \frac{q^2}{2C} \quad [\text{Potential Energy}]$$

$U = \frac{1}{2} C V^2$

 (potential energy).

⚡ The potential energy of a charged capacitor may be viewed stored electric field between plates

Energy Density

In a parallel-plate capacitor, neglecting fringing, the electric field has the same value at all points between the plates. Energy density $u \rightarrow$ Potential energy per unit volume between the plates — should be uniform.

$$u = \frac{U}{Ad} = \frac{CV^2}{2Ad} \quad \left[U = \frac{1}{2} CV^2 \right]$$

$$= \frac{1}{2} \epsilon_0 \left(\frac{V}{d} \right)^2 \quad \left[C = \frac{\epsilon_0 A}{d} \right]$$

As, $E = \frac{-\Delta V}{\Delta s} = \frac{V}{d}$ equals the electric field magnitude E , so $\boxed{u = \frac{1}{2} \epsilon_0 E^2}$ (energy density).

Sample problem 25.04 ①

An isolated conducting sphere whose radius R is 6.85 cm has a charge $q = 1.25 \text{ nC}$

(a) How much potential energy is stored in the electric field of this charged conductor?

(b) What is the energy density at the surface of the sphere.

① isolated sphere has capacitance $C = 4\pi\epsilon_0 R$ and energy U stored in a capacitor depends —

$$U = \frac{q^2}{2C} = \frac{q^2}{8\pi\epsilon_0 R} = \frac{C^2}{(F/m)(m)} = \text{nJ}$$

② $U = \frac{1}{2} \epsilon_0 E^2$, $E = \frac{1}{4\pi\epsilon_0} \frac{q}{R^2}$

$$U = \frac{1}{2} \epsilon_0 \left(\frac{1}{4\pi\epsilon_0} \frac{q}{R^2} \right)^2 = \frac{C^2}{C^2/N.m^2 \cdot m^4} = \frac{\mu J}{m^3}$$

Capacitor with Dielectric

key notes: ① Identify that capacitance is increased if the space between the plates is filled with a dielectric material.

② If space filled with a given dielectric material having dielectric constant k , we identify that ϵ_0 are modified by $k\epsilon_0$

③ Distinguish polar dielectrics from non polar dielectrics.

Dielectric: An insulating material — mineral or plastic.

→ Credit to Michael Faraday → who found that if we put dielectric material inside a capacitor, the ~~dielectric~~ capacitance increased by a factor k , which he called the 'dielectric constant' of insulating materials.

The dielectric constant of vacuum is unity by definition.

⊙ Another effect of the introduction of a dielectric is to limit the potential difference that can be applied between the plates to a certain value V_{max} (breakdown potential)

If this value is substantially exceeded, the dielectric material will breakdown and form a conducting path between the plates.

→ Every dielectric material has a characteristics "dielectric strength" which is the maximum value of the electric field that it can tolerate without breakdown.

⑧

As we discussed, the capacitance of any capacitor can be written in the form $C = \epsilon_0 \kappa \frac{A}{d}$ ($\frac{A}{d}$ has the dimension of length). For example, $\frac{A}{d} = A/d$, for a parallel-plate capacitor. $\rightarrow$ with a dielectric completely filling the space between the plates becomes $\rightarrow$

$$C = \kappa \epsilon_0 \frac{A}{d} = \kappa C_{\text{air}} \quad \left| \begin{array}{l} C_{\text{air}} \rightarrow \text{when capacitor} \\ \text{space is filled with} \\ \text{air} \end{array} \right.$$

⑨ In a region completely filled by a dielectric material of dielectric constant κ , all electrostatic equations containing the permittivity constant ϵ_0 are to be modified by replacing ϵ_0 with $\boxed{\kappa \epsilon_0}$

$$E = \frac{1}{4\pi \kappa \epsilon_0} \frac{q}{r^2}$$

$$E = \frac{\sigma}{\kappa \epsilon_0} \quad [\kappa \text{ is always greater than unity}]$$

both these equations show that for a fixed distribution of charges the effect of a dielectric is to weaken the electric field that would otherwise be present.

Work and energy when a dielectric is inserted into a capacitor.



The charging battery is now disconnected, and a porcelain slab ($k = 6.50$) is slipped between the plates.

(a) What is the potential energy of the capacitor before the slab is inserted?

$$U_i = \frac{1}{2} C V^2 = \frac{q^2}{2C} \\ = \frac{1}{2} (13.5 \times 10^{-12} \text{ F}) (12.5 \text{ V})^2 \approx 1055 \text{ pJ}$$

(b) What is the potential energy of the capacitor-slab device after the slab is inserted?

$$U_f = \frac{q^2}{2kC} = \frac{U_i}{k} = \frac{1055 \text{ pJ}}{6.50} \approx 162 \text{ pJ} \approx 160 \text{ pJ}$$

The missing energy

$$W = U_i - U_f = (1055 - 162) \text{ pJ} = 893 \text{ pJ}$$

The missing energy, in principle, ~~would be apparent to the person who introduced the slab.~~ The capacitor, ~~would be apparent to~~ exert a tiny tug on the slab and do work on it, in amount —

$$W = U_i - U_f = (1055 - 162)$$

If the slab were allowed to slide between the plates with no restraint and if there were no friction, the slab would oscillate back and forth between the plates with a mechanical energy of 893 pJ.

→

Dielectrics and Gauss law

We enclose the charge q on the top plate with a Gaussian surface from

Gauss law, we find

$$\epsilon_0 \oint \vec{E} \cdot d\vec{A} = \epsilon_0 EA = q$$

$$E_0 = \frac{q}{\epsilon_0 A} \quad \text{--- (i)}$$

but it now encloses the induced charge $-q'$ on the top face of the dielectric

$\rightarrow +q$ is free charge but $-q'$ is not free charge.

The net charge enclosed by the Gaussian surface is $q - q'$

$$\boxed{E = \frac{q - q'}{\epsilon_0 A}}$$

--- (ii)

Comparing

$$\epsilon_0 \oint \vec{E} \cdot d\vec{A} = q - q'$$

$$\boxed{E = \frac{q - q'}{\epsilon_0 A}}$$

--- (iii)

The effect of the dielectric is to weaken the original field E_0 by a factor of κ , so we may write,

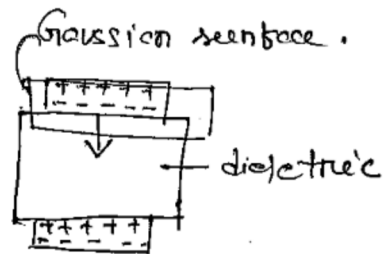
$$E = \frac{E_0}{\kappa} = \frac{q}{\kappa \epsilon_0 A} \quad \text{--- (iv)}$$

Comparing (ii) and (iv) $q - q' = \frac{q}{\kappa}$

$$\epsilon_0 \oint \vec{E} \cdot d\vec{A} = \frac{q}{\kappa}$$

$$\boxed{q = \epsilon_0 \oint \kappa \vec{E} \cdot d\vec{A}}$$

Gauss law with dielectric.



- Note:
1. The flux integral now involves $\vec{kE}$, not just $\vec{E}$.
 2. The vector $\boxed{\epsilon_0 k \vec{E}}$ is sometimes called the electric displacement $\vec{D}$, $\rightarrow \oint \vec{D} \cdot d\vec{A} = q$.

Sample problem 25.06

$$A = 115 \text{ cm}^2$$

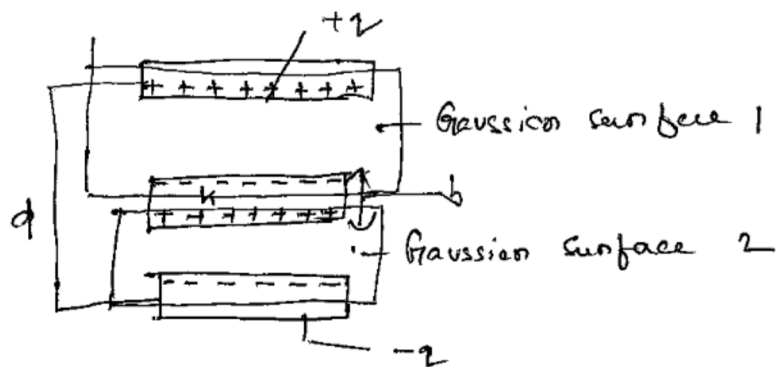
$$d = 1.24 \text{ cm}$$

$$V_0 = 85.5 \text{ V}$$

$$b = 0.780 \text{ cm}$$

$$k = 2.61$$

$$C_0 = \frac{\epsilon_0 A}{d} = \frac{(8.85 \times 10^{-12})}{1.24 \times 10^{-2} \text{ m}}$$



$\rightarrow$ (a) What is the capacitance C_0 before the dielectric slab is inserted?

$\rightarrow$ (b) What free charge appears on the plates?

$\rightarrow$ (c) What is the electric field E_0 in the gaps between the plates and the dielectric slab?

$\rightarrow$ (d) What is the electric field E_1 in the dielectric slab?

$\rightarrow$ (e) What is the potential difference V between the plates after the slab has been introduced?

$\rightarrow$ (f) What is the capacitance with the slab in place?